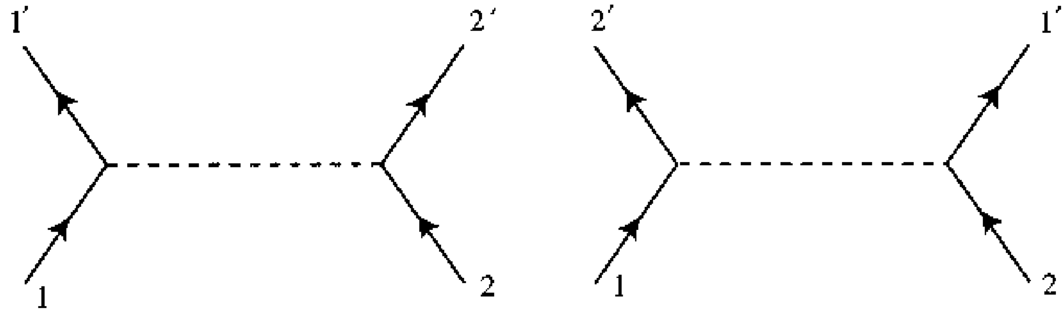

Figure 6.4. The connected second-order diagrams for fermion–fermion scattering in a theory with interaction (6.1.18). Here straight lines represent fermions; dotted lines are neutral bosons. There is a minus sign difference in the contributions of these two diagrams, arising from an extra interchange of fermion operators in the pairings represented by the second diagram.

term in Eq. (6.1.1) appear in the order:

$$a(2'^c)a(1')\psi^\dagger(x)\psi(x)\psi^\dagger(y)\psi(y)a^\dagger(1)a^\dagger(2^c)\ . \tag{6.1.22}$$

Here again there are two Feynman diagrams to this order, corresponding to the pairings

$$[a(2'^c)\psi(x)]\ [a(1')\psi^\dagger(x)]\ [\psi(y)a^\dagger(1)]\ [\psi^\dagger(y)a^\dagger(2^c)] \tag{6.1.23}$$

and

$$[a(2'^c)\psi(x)]\ [a(1')\psi^\dagger(y)]\ [\psi(y)a^\dagger(1)]\ [\psi^\dagger(x)a^\dagger(2^c)]\ . \tag{6.1.24}$$

(See Figure 6.5.) To go from (6.1.22) to (6.1.23) requires an even permutation of fermionic operators (for instance, move $\psi(x)$ past two operators to the left and move $\psi^\dagger(y)$ past two operators to the right) so there is no extra minus sign in the contribution of the pairing (6.1.23). On the other hand, to go from (6.1.22) to (6.1.24) requires an odd permutation of fermionic operators (the same as for (6.1.23), *plus* the interchange of $\psi^\dagger(x)$ and $\psi^\dagger(y)$) so the contribution of this pairing does come with an extra minus sign.**

Additional signs are encountered when we consider contributions of higher order. In theories of the type considered here, in which the interactions of fermions all take the form (6.1.18), the fermion lines in

** Actually, this sign is not wholly unrelated to the requirements of Fermi statistics. The same field can destroy a particle and create an antiparticle, so there is a relation, known as 'crossing symmetry', between processes in which initial particles or antiparticles are exchanged with final antiparticles or particles. In particular the amplitudes for the process $1\,2^c \to 1'2'^c$ are related to those for the 'crossed' process $1\,2' \to 1'2$; the two pairings (6.1.23) and (6.1.24) just correspond to the two diagrams for this process, which differ by an interchange of 1 and 2' (or 1' and 2), so the antisymmetry of the scattering amplitude under interchange of initial (or final) particles naturally requires a minus sign in the relative contribution of these two pairings. However, crossing symmetry is not an ordinary symmetry (it involves an analytic continuation in kinematic variables) and it is difficult to use it with any precision for general processes.